Sebastián Rueda Parra

518 246-0936 sebastianruedaparra13@gmail.com in/srpns

Summary

Data Scientist and Engineer with experience in neuroscience research, integrating robotics, hardware, AI, and advanced analytics to study human movement and sensation. Skilled in full-stack development—from hardware and firmware to data modeling and AI-based analysis—with the goal of extracting insights that inform clinical decisions. Curious, driven, and adaptable, thriving on learning and solving complex challenges.

Education

PhD in Electrical Engineering. University of Idaho. Moscow, Idaho (Dec 2022)
Bachelors in science. Engineering in Electronics. Universidad del Valle, Colombia (June 2016)

Skills:

- **Programming & Software:** Python, Matlab, C, C++, Arduino C , R, Java, SQL, VHDL
- **AI & Machine Learning:** Deep Learning, TensorFlow, PyTorch, LLMs, AI assistants (ChatGPT, DALL·E, Co-pilot, Claude), and Agents (OpenAI Agent SDK, CrewAI).
- **Data Analysis & Modeling:** Data Visualization, Inferential Statistics, Estimation Theory, Statistical Digital Signal Processing, Causality Estimation
- **Tools & Platforms:** Git, Anaconda, LaTeX, Windows, Linux, Microsoft Office, Inkscape
- **Professional & Interpersonal:** Interdisciplinary teamwork, Mentoring & Teaching, Effective communication, Critical thinking, Motivated, Quick learner
- **Research & Scientific Skills:** Experimental Design, Data Processing, Hypothesis Testing, Manuscript Preparation, Scientific Writing, Public Speaking
- **Languages:** Spanish (Native), English (Professional proficiency), German (Advanced)
- **Hobbies & Interests:** Piano, Sketching, Table Tennis

Research Experience

National Center for Adaptive Neurotechnologies (NCAN) **Albany, NY**
Postdoctoral Researcher *January 2023 – present*

- Conducted research on robotics and bio-signals to study sensorimotor function in U.S. veterans with stroke, Parkinson's, and other neurological conditions.
- Evaluated test–retest reliability of Electroencephalography (EEG) and behavioral biomarkers for assessing motor function and proprioception.
- Implemented BCI-based assessments for detection of color vision deficits (CVD).
- Classified congenital red-green CVDs, based on EEG data with 93 % accuracy.
- Optimized BCI-based CVD classification, reducing testing time from 40 minutes to 1 minute
- Applied DL models for EEG feature extraction related to proprioceptive processing.
- Characterized cutaneous reflexes for phantom limb pain therapy.
- Contributed to grant writing for federally funded neuroscience and rehabilitation projects.

University of Idaho. **Moscow, Idaho.**
Research Assistant *July 2017—Dec 2022*

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- Studied proprioception using kinematics, robotic tools and electrophysiology.
- Contributed to the realisation of a bilateral robotic exoskeleton for motor assessment and rehabilitation.
- Applied AI and computer vision to detect plant diseases from images with 90% accuracy.
- Analyzed bioelectrical signals (EEG, EMG) to investigate neural and muscular contributions to motor control and sensation.
- Designed experimental setups for biomedical research hypothesis testing.
- Applied advanced visualization techniques to interpret multidimensional physiological data from individuals with stroke.

Universidad del Valle. **Cali, Colombia.**
Research Assistant *Sept 2014 – Jun 2016*

- Designed and implemented mobile robots for research and educational applications.
- Investigated technology-driven strategies to teach algorithmic thinking in students.

Professional Experience

National Center for Adaptive Neurotechnologies (NCAN) **Albany, NY**
Postdoctoral Researcher *January 2023 – present*

- Used AI assistants and intelligent agents to streamline workflows and automate tasks.
- Built automated analysis and visualization pipelines, improving project efficiency.
- Worked with cross-functional international teams on data-driven healthcare solutions.
- Developed visualization tools enabling non-technical individuals to interpret complex data.
- Developed software tools that enabled non-technical individuals to deliver experiments.
- Designed the controls of a novel BCI-based visual stimulator for assessing Color Vision.
- Designed and developed hardware components, including PCB layouts and prototypes.

University of Idaho. **Moscow, Idaho.**
PhD Student *July 2017—Dec 2022***.**

- Analyzed EEG/EMG signals to reveal neural and muscular contributions to motor function.
- Produced documentation for complex hardware and software.
- Built custom circuits and experimental setups to support biomedical research protocols.
- Developed tools that enabled healthcare individuals to deliver experiments.
- Created visualization tools to interpret multidimensional physiological datasets.
- Ensured proper timing synchronization of biosignal acquisition in Robotic applications.

Lynks Ingeniería **Cali, Colombia**
Research and Development Engineer *Jan 2016—June 2017.*

- Full-stack developer (hardware, firmware, software) for farming monitoring products.
- Delivered advanced technical support addressing a broad range of client issues.
- Developed low-power firmware for 8-bit microcontrollers, improving battery life by 500%.

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Fraunhofer IWES institute. **Kassel, Germany**
Data-Science Intern. *Jan – July 2014*

- Gained foundational knowledge in neural networks and their applications.
- Collaborated with interdisciplinary and multicultural teams to deliver project goals.
- Developed an ANN-based regression for electric load forecasting with 88% accuracy.

Teaching Experience

National Center for Adaptive Neurotechnologies (NCAN) **Albany, NY**
Postdoctoral Researcher *January 2023 – present*

- Delivered presentations and hands-on activities during research summer camps.
- Mentored interns in PCB design and electrophysiological data analysis.

University of Idaho. **Moscow, Idaho.**
Teaching Assistant *July 2017—Dec 2022*

- Mentored undergraduate students in biomedical experiments, data analysis, and research.
- Assisted in teaching Electrical Circuits and Digital Logic Laboratory courses in EE department.

Self-employed **Cali, Colombia**
Informal Instructor *2010 – 2012*

- Provided one-on-one instruction in basic programming (Java), Calculus, and Linear Algebra.

Contributions to science:

Articles

Rueda, S., et al. (2025) BCI-based detection and classification of congenital CV deficiencies. (*In preparation*).

Hardesty R, **Rueda, S.**, et al. (2025) Characterization of cutaneous reflex symmetry elicited in the saphenous nerve. (*In preparation*).

Wang L, Zadehgol A, R, **Rueda, S.**, et al. (2025) A multimodal deep learning framework for proprioceptive error prediction. (*In preparation*).

Rueda, S., et al. (2025) Test-Retest Reliability of Kinematic and EEG Low-beta spectral features in a robot-based arm movement task. (*Biomedical Physics & Engineering Express*).

Rueda, S., Perry, J., Wolbrecht, E., Reinkensmeyer, D., & Gupta, D. (2024). Multidimensional feature analysis shows stratification in robotic-motor-training gains based on the level of pre-training motor impairment in stroke (*IEEE-EMBC*).

Rueda, S., Perry, J., Wolbrecht, E. & Gupta, D. (2024). Neural correlates of Bilateral Proprioception and Adaptation with Training (*PLOS ONE*).

Bitikofer, C., **Rueda, S**., Maura, R., Wolbrecht, E., Perry, J. (2024) BLUE SABINO: Development of a BiLateral Upper-Limb Exoskeleton for Simultaneous Assessment of Biomechanical and Neuromuscular Output (*Machines MDPI*).

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Maura, R., **Rueda, S**., Stevens, R., Weeks, D., Wolbrecht, E., & Perry, J. (2023). Literature Review of Sensor-Based Upper Extremity Stroke Assessment via EEG, EMG, Kinematic and Kinetic Measurements and Their Reliability (*JNER*).

Cortes, E. B. B., Gaviria, B. F., & **Rueda, S**. (2017). Development of a set of mobile robots for basic programming experimentation (*Revista UIS Ingenierías*).

Doctoral Dissertation

Rueda S. (2022). Electrophysiological Data Analysis to Study Human Motion and Sensation: Searching for Features that Adapt with Training or Injury. University of Idaho Library Digital Collections.

Abstracts and Posters with Oral Presentation

Rueda, S., et al. (2025) Improving validation of BCI-based Color Vision assessment, BCI society.

Rueda, S., et al. (2024). Contralateral Parietal Low-Beta Power informs of Proprioceptive Uncertainty in Chronic Stroke., SfN.

Rueda, S., et al. (2024). Reliability testing of EEG spectral features in a robot-based arm movement task., SfN.

Rueda, S., et al. (2022). Pre-training neural correlates for predicting gains from robot-assisted finger training after stroke. 2022, ASNR.

Rueda, S., et al. (2022). Neural Correlates of Bilateral and Unilateral Proprioception in People with Musical Instrument Training, SfN.

Rueda, S., et al. (2021). Visualization of multivariate behavioral data in stroke subjects during robotic rehabilitation therapy for fingers, SfN.

Ketkar, V., **Rueda, S.**, et al. (2021). Design and Development of a Spherical Five-Bar Thumb Exoskeleton Mechanism for Post-Stroke Rehabilitation, ICORR.

Rueda, S., Wolbrecht, E., & Perry, J (2019). Characterization of movement artifacts in EEG measurements toward robot-aided neuromuscular assessment, ASNR.

Honors

2020	Outstanding Teaching assistant, University of Idaho
2013	DAAD young Engineer scholarship holder
2010, 2011, 2012, 2015	Top 5 best GPA scores, Universidad del Valle

Memberships and affiliations

IEEE Engineering in Medicine and Biology Society (EMBS)	2024 – 2025
BCI Society	2024 – 2025
Society for Neuroscience (SfN)	2019 – 2025
American Society for NeuroRehabilitation (ASNR)	2019 – 2023